D 11759 **(Pages : 4)** **Name.....**

Reg. No.....

THIRD SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2021

B.B.A.

BBA 3A 11—BASIC NUMERICAL SKILLS

(2014—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A

Answer **all** *questions.*
Each question carries 1 *mark.*

I. Choose the correct answer :

1 The set with no element is denoted by ————————.

a) 0. b) ø.

c) A. d) $.

2 If A and B are disjoint sets, then A ∩ B is ————————.

a) 0. b) A.

c) B. d) ø.

3 Which of the following measure possesses mathematical properties ?

a) AM. b) GM.

c) HM. d) All of these.

4 Index Number is a ————————.

a) Measure of relative changes. b) A special type of an average.

c) A percentage relative. d) All the above.

5 Skewness refers to ————————.

a) Flatness. b) Peakedness.

c) Symmetry. d) Asymmetry.

Turn over

II. Fill in the blanks :

6 Find mode 23, 35, 28, 42, 62, 53, 35, 28, 42, 35, 23, 42, 3 ————————.

7 A matrix in which every element is zero is called ————————.

8 Row headings of a table is called ————————.

9 Find the 15th terms of the sequence 20, 15, 10 ————————.

10 The solution of the equation $4 = \frac{2}{3} \times$ ————————.

(10 × 1 = 10 marks)

Part B

Answer any **eight** *questions.*
Each question carries 2 *marks.*

11 If $A = \{a, b, c, d\}$; $B = \{d, e, f, g\}$; $C = \{h, i, j, k\}$. Prove that $(A - B) \cap (A - C) = A - (B \cup C)$.

12 Represent $A \cap B'$ by using Venn diagram.

13 If $A = \{1, 2\}$; $B = \{4, 5\}$. Find $B \times A$.

14 Find $2A - 3B$

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & -1 & 5 \end{pmatrix}; B = \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix}.$$

15 Find the value of determinants :

$$A = \begin{vmatrix} 1 & 5 & 2 \\ 3 & 1 & 2 \\ 6 & 2 & 5 \end{vmatrix}.$$

16 Solve the equation $2x^2 + 8x + 8 = 0$.

17 Solve $2x - y = 5$
$3x - 4y = 10$.

18 In an arithmetic progression the sum of the first 10 terms is 400 and the sum of the next ten terms is 1000. Find the common difference and the first term.

19 Find the number of years a sum of Rs. 10,000 will take to become 19,600 if the rate of interest is 8 %.

20 What do you mean by cartogram ?

(8 × 2 = 16 marks)

Part C

Answer any **six** *questions.*
Each question carries 4 *marks.*

21 Find Karl Pearson's co-efficient of skewness for the given values :

25, 18, 32, 20, 25, 48, 72, 24, 50, 25

22 Describe the steps in Statistical investigation.

23 With Median as base calculate Mean Deviation and compare the variability of the two series :

Series A : 3484, 4572, 4124, 3682, 5624, 4388, 3680, 4308

Series B : 487, 508, 620, 382, 408, 266, 186, 218

24 State the limitations of statistics.

25 Solve the following equation by using matrices :

$2x - 3y = 3$
$4x - y = 11.$

26 Draw appropriate Venn diagram for each of the following :

$(A \cup B)^c$ $A \cap B^c \cap C^c$

27 Find the compound interest for Rs. 7,000 for 4 years if interest is payable half yearly at 6 % per annum.

28 A bank offers 5 % compound interest calculated on half-yearly basis. A customer deposits Rs. 1,600 each on 1st January and 1st July of a year. Find the interest he would have gained at the end of the year.

(6 × 4 = 24 marks)

Turn over

Part D (Essay Questions)

Answer any **two** *questions.*
Each question carries 15 *marks.*

29 Each student in a class of 40, studied at least one of the languages Malayalam, Hindi and French. 16 studied Malayalam, 22 French, 26 Hindi, 5 studied Malayalam and French, 14 Hindi and French, 2 Malayalam, Hindi and French. Find the number of students who studied Malayalam and Hindi but not French.

30 Solve the following equations by using matrices :

$$3x + 2y + z = 6$$
$$2x - 3y + 3z = 2$$
$$x + y + z = 3.$$

31 Calculate standard deviation and co-efficient of variation from the following data :

Class	:	0–5	5–10	10–15	15–20	20–25	25–30
Frequency	:	6	8	10	15	12	19

(2 × 15 = 30 marks)